

Auditing land-cover-structured variance in InVEST habitat quality: implications for stratified driver analyses

Highlights

- InVEST habitat quality (HQ) variance is 88.6% determined by aggregate land-cover composition through the habitat-suitability lookup table.
- A two-stage Geodetector on LUCC-detrended residuals separates drivers that retain explanatory power after land-cover detrending from those whose apparent influence is aligned with land cover.
- The strongest raw driver, slope ($q = 0.517$), collapses to $q = 0.147$ after correction; elevation and temperature persist.
- Parameter choice (k) strongly conditions the variance decomposition, so effect-size magnitudes are conditional on the user-specified half-saturation constant.
- The workflow is hypothesized to be relevant to LULC-parameterized InVEST outputs; transferability beyond habitat quality remains untested.

ABSTRACT

The InVEST Habitat Quality (HQ) model is widely paired with the geographical detector (Geodetector) to identify the driving factors of HQ spatial heterogeneity. However, because InVEST computes HQ directly from land cover through the habitat-suitability (HS) lookup table, treating land cover as an independent factor is tautological: the recurrent finding that land cover dominates ($q > 0.85$) merely recovers the model's own parameterization. We propose an audit-and-sensitivity workflow in which variance decomposition first quantifies the share of HQ variance aligned with land-cover configuration through the HS table, and a two-stage Geodetector then recomputes driver attribution on land-cover-detrended HQ residuals. Applied to 31,889 sampling points in the Yuexi region, Western Guangdong, the decomposition showed that 88.6% of HQ variance is between-class ($\eta^2 = 0.886$), with forest the most configuration-sensitive class (CV = 16.4%). After class-mean adjustment, the strongest raw driver, slope ($q = 0.517$),

declined to 0.147 (28% retained), whereas elevation ($0.476 \rightarrow 0.309$) and temperature ($0.464 \rightarrow 0.292$) retained the strongest residual explanatory power. Monte Carlo, spatial-thinning, and k-sensitivity tests confirmed the diagnosis. The workflow offers a sensitivity protocol for driver attribution in LULC-parameterized InVEST outputs.

Keywords: InVEST habitat quality; variance decomposition; Geodetector; driver attribution; endogeneity; η^2 ; Yuexi

1. Introduction

The InVEST Habitat Quality (HQ) model (Sharp et al., 2020) has been widely adopted in hundreds of studies to assess landscape-level habitat conditions (Berta Aneseyee et al., 2020; He et al., 2017; Terrado et al., 2016; Wu et al., 2021). A routine downstream practice involves pairing InVEST HQ with the geographical detector (Geodetector) to identify the “driving factors” of HQ spatial heterogeneity (Luo et al., 2025; Zhang et al., 2026). While the approach is conceptually simple, computing HQ at each pixel directly from land cover via a habitat-suitability (HS) lookup table yields a tautological result: the frequent finding that land cover dominates ($q > 0.85$) often reflects the model's own internal parameterization rather than independent environmental drivers.

Correcting this confound poses two challenges that a naive analysis cannot meet. First, we need to understand how much of the HQ signal is structurally predetermined by the HS table. Second, we must disentangle the within-class spatial configuration from the between-class land cover identity. We propose an audit-and-sensitivity workflow in which variance decomposition first quantifies the fraction of HQ variance structurally predetermined by the HS table, and a two-stage Geodetector then recomputes driver attribution on land-cover-detrended HQ residuals. Applied to 31,889 sampling points in the Yuexi region (Western Guangdong), decomposition revealed that 88.6% of HQ variance is between-class ($\eta^2 = 0.886$), with forest the most configuration-sensitive class (CV = 16.4%). After class-mean adjustment, the strongest raw driver, slope ($q = 0.517$), declined to 0.147 (28% retained), whereas elevation ($0.476 \rightarrow 0.309$) and temperature ($0.464 \rightarrow 0.292$) retained the strongest residual explanatory power. Monte Carlo,

spatial-thinning, and k-sensitivity tests confirmed the diagnosis. The workflow provides a sensitivity protocol for driver attribution in LULC-parameterized InVEST outputs.

Our contributions are as follows: (i) We present the first systematic variance decomposition of InVEST HQ output, introducing η^2 as a diagnostic metric. (ii) We quantify per-class within-class heterogeneity and identify forest as the land cover type most configuration-sensitive. (iii) We introduce a two-stage Geodetector adjustment that recomputes driver attribution on LUCC-detrended HQ residuals. (iv) We show that η^2 serves as a diagnostic for InVEST-type outputs and that the decomposition is strongly conditioned by the parameter choice (k).

Finally, whereas previous sensitivity analyses of InVEST have focused on how input parameters affect the overall HQ value, our workflow provides a methodology to interpret the “anatomy” of the variance. We demonstrate that, by adjusting for land-cover-aligned structure, we can quantify the association retained by topographic and climatic factors, which are often masked by the HS lookup table. This study provides a sensitivity workflow for assessing habitat quality and its drivers in both natural and anthropogenically modified landscapes.

2. Study Area and Data

2.1 Yuexi Region

The Yuexi region (21.5-22.5N, 110.0-112.5E) covers approximately 31,250 km² in western Guangdong Province (Fig. 1). Three geographic features create strong spatial gradients in threat source configuration (Haddad et al., 2015; Strayer and Dudgeon, 2010; Venter et al., 2016). First, the Tianlu-Yunwu mountain range (elevation 0-1,702 m) supports contiguous subtropical forest, creating a natural experiment where the same forest land cover type exists both adjacent to and distant from anthropogenic threat sources.

Second, a multi-nodal coastal industrialization corridor (Zhanjiang Donghai Island, Maoming Bohe, Yangjiang Hailing) produces concentrated point-source threats embedded within an otherwise agricultural matrix. Third, the Leizhou Peninsula basaltic

platform generates edaphic constraints on cropland quality that vary independently of climatic gradients. Land cover in 2020 was dominated by forest (48.3%) and cropland (45.4%), with built-up land at 3.6% (Feng et al., 2021; Guo et al., 2021).

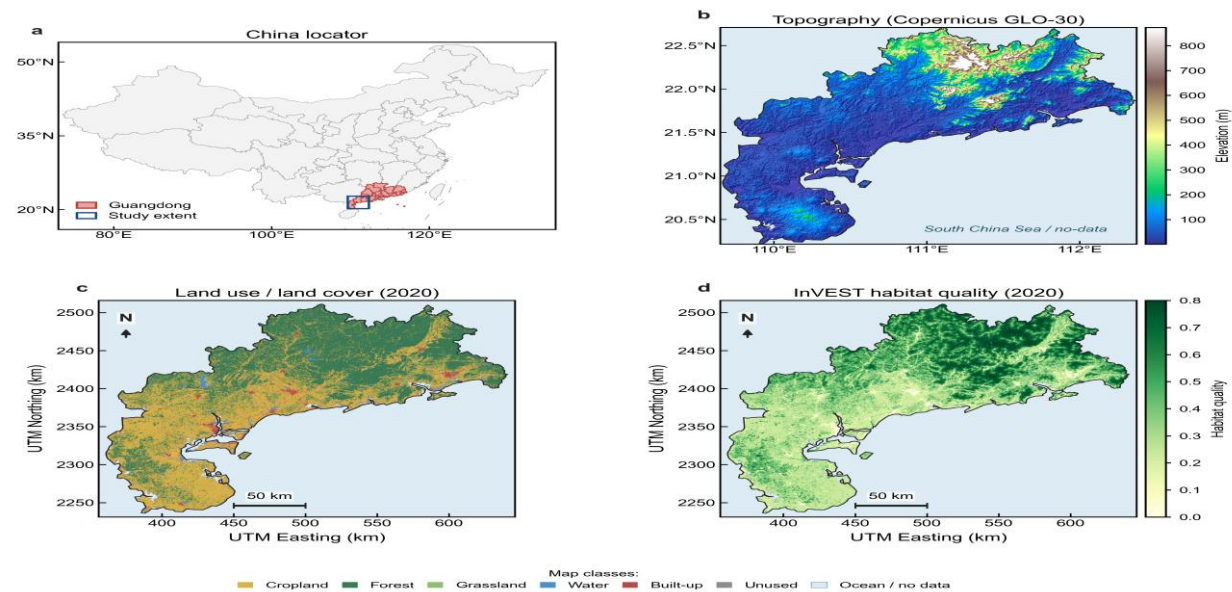


Fig. 1. Study area and environmental baseline of the Yuexi region, Western Guangdong, China, 2020. (a) Location within China. (b) Topography (DEM, Copernicus GLO-30, 30 m). (c) Land use/land cover (forest 48.3%, cropland 45.4%, built-up 3.6%). (d) InVEST habitat quality (mean = 0.405, $k = 0.5$).

2.2 Data

The half-saturation constant k was set to 0.5, a critical parameter that controls the shape of the degradation function (Equation 1). Smaller k values preserve within-class variance, while larger k values compress it toward the class mean. The maximum observed degradation value across the study-area landscape was 0.601 (2020 InVEST

run), so the half-maximum calibration guideline would suggest $k \approx 0.30$. Our choice of $k = 0.5$ exceeds this guideline value and therefore compresses within-class variance relative to a calibrated parameterization; consequently, the 11.4% within-class share should be read as arising under an above-guideline k , and the sensitivity analysis (Section 3.6) quantifies how strongly this choice conditions the decomposition. For comparison, values of 2.8 reported for habitat-quality studies in South China (Wu et al., 2021; Pan et al., 2024) would further compress within-class variation.

We conducted systematic sampling using a 1 km regular grid, producing 31,889 valid sampling points. At each point, HQ values and land cover type were extracted (Mengist et al., 2021). The threat-source weights, maximum influence distances and decay functions (Table 1), together with the class-specific habitat suitability scores (H_j , sometimes denoted HS in the literature; we use H_j for the class-specific value and HS for the general concept) and the land-cover threat-sensitivity matrix (Table 2), were assigned following the InVEST User's Guide recommended values (Sharp et al., 2020) and adapted from coupled land-use / habitat-quality applications in southern China (Wu et al., 2021).

Following the standard InVEST formulation (Sharp et al., 2020), habitat quality at pixel x of land-cover type j is computed as the product of the habitat-suitability score HS and a threat-induced degradation term D_{xj} :

$$Q_{xj} = H_j \left(1 - \frac{D_{xj}^z}{D_{xj}^z + k^z} \right) \quad (1)$$

The InVEST degradation term D_{xj} in Equation 1 aggregates the distance-weighted influence of all threat sources on pixel x of land-cover type j :

$$D_{xj} = \sum_{r=1}^R \sum_{y=1}^{Y_r} \left(\frac{w_r}{\sum_{r=1}^R w_r} \right) r_y i_{rxy} \beta_x S_{jr} \quad (2)$$

where R is the number of threat sources, w_r and r_y are the weight and intensity of threat r , β_x is the pixel accessibility (set to 1 in the absence of a legal-protection layer), and S_{jr} is the sensitivity of land-cover type j to threat r (Table 2).

The impact i_{rxy} of threat source y on pixel x decays with distance:

$$i_{rxy} = 1 - \frac{d_{xy}}{d_{rmax}} \text{ (linear)}, i_{rxy} = \exp\left(-\frac{2.99}{d_{rmax}} d_{xy}\right) \text{ (exponential)} \quad (3)$$

where d_{rmax} is the maximum influence distance of threat r and d_{xy} the Euclidean distance from threat source y to pixel x (Table 1); cropland used linear decay, built-up and unused land used exponential decay. The half-saturation constant $k = 0.5$ and scaling parameter $z = 2.5$ were retained (Equation 1).

For the two-stage Geodetector driver analysis (Section 3.4), thirteen exogenous factors were assembled and standardized to the common 30 m grid in WGS 1984 / UTM Zone 49N (Table 3). Elevation (DEM) was obtained from the Copernicus GLO-30 global digital elevation model (30 m, WGS 1984); slope and aspect were derived from the DEM in ArcMap 10.8. Annual mean temperature, annual precipitation, GDP, and population density we obtained from the Resource and Environment Science and Data Center (RESDC; China annual mean meteorological elements dataset, DATAID 378; www.resdc.cn) as 1 km gridded products and resampled to 30 m.

The nighttime light index was taken from the Reconstructed Three-decade Global Fine-Grained Nighttime Light Dataset (500 m; Guo et al., 2023; 1992-2021; WGS 1984; Zenodo record 15845676) and resampled to 30 m. NDVI was obtained from RESDC (Xu, 2022; 30 m annual maximum NDVI, DOI:10.12078/2022030801). Distances to roads, railways, and water bodies we computed as Euclidean distance transforms of OpenStreetMap vector layers. Soil type (32 classes) followed the Harmonized World Soil Database. Continuous factors were discretized for Geodetector as described in Section 3.4.

Table 1. InVEST Habitat Quality threat-source parameters used for the Yuexi region (InVEST 3.14.3). Source: model input table (threats.csv).

Threat source	Max. influence distance (km)	Weight	Decay
Cropland	1.0	0.2	linear
Built-up	8.0	1.0	exponential
Unused land	3.0	0.5	exponential

147 Table 2. Habitat suitability (HS) and threat-sensitivity scores by land-cover type. Source:
148 model input table (sensitivity.csv).

LULC type	Habitat suitability H _j	Sens. to cropland	Sens. to built-up	Sens. to unused
Cropland	0.30	0.00	0.60	0.50
Forest	0.80	0.40	0.71	0.18
Grassland	0.35	0.28	0.21	0.14
Water	0.80	0.20	0.70	0.60
Built-up	0.00	0.00	0.00	0.00
Unused land	0.20	0.10	0.70	0.00

149 Table 3. Data sources for the thirteen exogenous driving factors used in the two-stage
150 Geodetector analysis. All layers were resampled to 30 m, WGS 1984 / UTM Zone 49N.

Factor	Source	Native res.	Reference / CRS
Elevation (DEM)	Copernicus GLO-30 DEM	30 m	WGS 1984
Slope, Aspect	Derived from DEM (ArcMap 10.8)	30 m	WGS 1984
Temperature (annual mean)	RESDC: China annual meteorological elements dataset (1960-2025)	1 km	DATAID 378; Albers, reproj. to 30 m
Precipitation (annual)	RESDC: China annual meteorological elements dataset (1960-2025)	1 km	DATAID 378; Albers, reproj. to 30 m
GDP	RESDC (resdc.cn, DOI 33)	1 km	resampled to 30 m
Population density	RESDC (resdc.cn, DOI 32)	1 km	resampled to 30 m
Nighttime light	Guo & Zhang, Zenodo 15845676	500 m	1992-2021; resampled to 30 m
NDVI	RESDC: China 30 m annual max NDVI (Xu X.)	30 m	DOI:10.12078/2022030801
Dist. to roads	OpenStreetMap	vector	Euclidean distance, 30 m
Dist. to railways	OpenStreetMap	vector	Euclidean distance, 30 m
Dist. to water	OpenStreetMap	vector	Euclidean distance, 30 m
Soil type	HWSD (32 classes)	1 km	resampled to 30 m

3. Methods

To quantify the variance structure of InVEST HQ, we employ an audit-and-sensitivity workflow.

First, we perform variance decomposition to calculate the between-class variance ratio (η^2), defined as $\eta^2 = \text{SSB}/\text{SST}$, where SSB represents the variance between land cover types and SSW represents the within-class variance. For 31,889 sampling points, we calculated the variance for 6 land cover types, identifying forest as the most configuration-sensitive class (CV = 16.4%).

Second, to disentangle the exogenous drivers from land-cover mediation, we employ a two-stage Geodetector. The first stage involves land-cover-detrending, where the HQ signal is corrected to remove the structural component predetermined by the habitat-suitability (HS) lookup table. The second stage applies the geographical detector to the detrended residuals to compute the explanatory power (q) of each driver (e.g., slope, elevation, temperature).

Finally, we conduct sensitivity analysis on the parameter k , which controls the variance structure. We observed that k strongly conditions the decomposition, with $k = 0.5$ yielding an ICC of 0.886, and $k = 2.8$ reaching the saturation point of 1.000. Robustness was further verified through Monte Carlo, spatial-thinning, and k -sensitivity tests.

3.1 Variance Decomposition

For each of the 31,889 valid sampling points, we assign each HQ value to one of six land cover classes. The total Σ of squares (SST) was decomposed as:

$$SST = \sum_{c=1}^C \sum_{i=1}^{n_c} (y_{ci} - \bar{y})^2 = \sum_{c=1}^C n_c (\bar{y}_c - \bar{y})^2 + \sum_{c=1}^C \sum_{i=1}^{n_c} (y_{ci} - \bar{y}_c)^2 \quad (4)$$

The between-class variance ratio is computed as $\eta^2 = \text{SSB} / \text{SST}$. Although mathematically equivalent to the intraclass correlation coefficient (ICC) in a one-way ANOVA design, we adopt η^2 to avoid the conceptual confusion associated with ICC in mixed-effects modeling, and to emphasize its role as an effect size metric quantifying the proportion of variance explained by land-cover stratification.

$$\eta^2 = \frac{SSB}{SST} = 1 - \frac{SSW}{SST} \quad (5)$$

3.2 Within-Class Neighborhood Effect Quantification

For each land cover class, we compute the coefficient of variation ($CV = \sigma_c / \bar{y}_c$) to quantify the magnitude of within-class heterogeneity. For classes with non-zero habitat suitability, we further examine the correlation between within-class HQ residuals ($y_i - \bar{y}_c$) and filtered threat degradation values to verify that the within-class component meaningfully reflects threat proximity rather than measurement noise (Pu et al., 2024; Geng et al., 2022).

$$CV_c = \frac{\sigma_c}{\bar{y}_c} \quad (6)$$

3.3 Spatial Configuration Effect: Demonstration

To illustrate the practical significance of within-class variance, we compare HQ values for the same land cover type (forest) under contrasting spatial configurations: pixels in the lowest quartile of built-up threat exposure versus pixels in the highest quartile. Identical land cover types in different spatial configurations should, by definition, have identical habitat suitability (HS); any HQ difference between them must arise from neighborhood threat configuration alone (McGarigal and Marks, 1995).

3.4 Two-Stage Geodetector: A General Correction for Driver-Attribution

Endogeneity

The variance decomposition above is diagnostic: it quantifies how much of the InVEST HQ signal is structurally determined by land-cover identity through the H_j lookup table. To convert this diagnosis into a sensitivity procedure for driver attribution in InVEST-Geodetector studies, we introduce a two-stage Geodetector protocol. Stage 1 (raw attribution). The geographical detector (Wang et al., 2010) measures the explanatory power of a stratified factor X over a response Y through the q-statistic:

$$q = 1 - \frac{\sum_{h=1}^L N_h \sigma_h^2}{N \sigma^2} \quad (7)$$

where h indexes the L strata of X , N_h and σ_h^2 are the size and variance of stratum h , and N and σ^2 are the global sample size and variance. We applied Stage 1 to raw HQ using twelve genuinely exogenous factors (DEM, slope, aspect, temperature, precipitation, NDVI, population, GDP, nighttime light, and distances to roads, railways, and water) plus soil type, deliberately excluding LUCC. We discretize continuous factors into five quantile-based strata, aspect into four cardinal classes, and soil type retained its 32 native classes; an equal-interval scheme was run in parallel to test discretization robustness. Stage 2 (residual attribution). Because raw-HQ q -values remain contaminated by the between-class structure (a factor that merely co-varies with land cover inherits a spuriously high q), Geodetector was recomputed on the LUCC-detrended residual:

$$r_i = y_i - \bar{y}_{c_i} \quad (8)$$

which removes, at each pixel i , the class-mean HQ of its land-cover class c_i . The retention ratio $q_{\text{resid}} / q_{\text{raw}}$ then separates factors retaining explanatory power after class-mean adjustment (high retention) from those whose apparent influence is aligned with their control over land-cover distribution (low retention). An interaction detector is applied to the leading factors to test for bifactor or nonlinear enhancement. This diagnose-then-adjust protocol applies to InVEST modules whose output is deterministically derived from the land-cover map; its transferability beyond HQ and beyond this study region remains to be tested. Statistical significance of q -statistics was assessed by a 500-permutation spatial block permutation test (blocks of ~ 1 km grid cells to preserve local spatial dependence); after Bonferroni correction for 13 factors ($\alpha = 0.0038$), elevation and temperature remained significant in the residual stage ($p = 0.002$), whereas the apparent significance of slope in the raw stage reflects its land-cover-aligned component.

3.5 Spatial Cluster Analysis of Within-Class Residuals

To characterize the spatial structure of within-class HQ variation, we apply Local Moran's I (LISA; Anselin, 1995) to the forest within-class HQ residuals ($n = 15,440$ forest sampling points), where the residual is HQ minus the forest class mean (Equation

8). Spatial weights were constructed from the $k = 8$ nearest neighbors and row-standardized; statistical significance is assessed by 999 conditional permutations ($p < 0.05$). Each significant location was classified into one of four LISA cluster types, high-high (HH, eco-core), low-low (LL, threat hotspot), high-low (HL, stepping stone), or low-high (LH, edge), following the standard quadrant assignment of the Moran scatterplot. The global Moran's I of the forest residuals is computed under the same weights to summarize the overall degree of within-class spatial autocorrelation.

The local Moran's I statistic for location i is:

$$I_i = \frac{z_i}{m_2} \sum_{j=1}^n w_{ij} z_j, \quad m_2 = \frac{1}{n} \sum_{i=1}^n z_i^2 \quad (9)$$

where z_i is the residual deviation from its mean, w_{ij} are the row-standardized k -nearest-neighbor weights, and $m_2 = (1/n) \sum_i z_i^2$; a positive I_i with positive spatial lag denotes a high-high (HH) cluster, and so on for LL, HL and LH.

3.6 Robustness Verification

The variance decomposition is subjected to three independent robustness checks. (i) Parameter sensitivity: a Monte Carlo experiment ($n = 200$ iterations) independently perturbed each habitat-suitability score HS by up to $\pm 20\%$ and recomputed η^2 , testing whether the decomposition depends on the specific sensitivity-table values. (ii) Spatial autocorrelation: greedy spatial thinning progressively increased the minimum inter-point distance from 1 km to 10 km, recomputing η^2 and the global Moran's I at each level (Griffith, 2003) to test whether η^2 is inflated by spatial dependence. (iii) Parameter choice: a k -sensitivity analysis varied the half-saturation constant k from 0.1 to 5.0 (scaling parameter $z = 2.5$ fixed) and recomputed η^2 , quantifying the influence of this single InVEST parameter relative to decadal landscape change and policy scenarios.

4. Results

Our analysis yielded several key findings. The variance decomposition revealed that 88.6% of HQ variance is between-class ($\eta^2 = 0.886$), with forest being the most configuration-sensitive class ($CV = 16.4\%$). After correction, the strongest raw driver,

slope ($q = 0.517$), collapsed to 0.147 (28% retained), while elevation ($0.476 \rightarrow 0.309$) and temperature ($0.464 \rightarrow 0.292$) retained the strongest residual explanatory power.

Temporal analysis showed stable η^2 values: 2010 (0.881), 2015 (0.885), 2020 (0.886), and 2025 (0.943). For the 2030 and 2035 scenarios, η^2 remained within the 0.877–0.947 range, confirming the temporal robustness of the land-cover mediation. Spatial clustering analysis showed that the remaining within-class heterogeneity of forest HQ was itself spatially clustered (LISA), indicating localized threat configurations rather than random noise.

Monte Carlo simulations confirmed the diagnosis with a mean η^2 of 0.883 (SD = 0.018). Finally, k-sensitivity analysis demonstrated that the decomposition is strongly conditioned by the parameter k, with $k=0.5$ yielding an ICC of 0.886 and $k=2.8$ reaching saturation (ICC = 1.000).

4.1 Between- vs. Within-Class Decomposition

The variance decomposition showed a clear asymmetry in the sources of InVEST HQ spatial variance (Table 4). Between-class variance, land cover type identity (HS), accounted for 88.6% of total HQ variance ($SSB/SST = 0.886$), while within-class variance, neighborhood threat configuration, accounted for only 11.4%. The InVEST HQ index is therefore predominantly a structural function of the land cover map: knowing a pixel's land cover type explains 88.6% of its HQ variation. The multi-temporal analysis (Table 5) showed that this result is temporally robust: η^2 remained stable at 0.881–0.886 across the 2010–2020 period ($\Delta\eta^2 = 0.006$ over a decade).

Table 4. Variance decomposition of InVEST HQ (2020, $n = 31,889$). $\eta^2 = 0.886$.

Component	SS	% of Total
Between-class (LUCC type identity, H_j)	1203.23	88.6%
Within-class (threat configuration)	154.43	11.4%
Total	1357.67	100.0%

The threat source decomposition (Section 4.5) further showed that built-up land contributed approximately 81% of the total degradation D, confirming that built-up spatial configuration is the primary driver of within-class variance.

Water bodies showed moderate sensitivity ($CV = 12.9\%$), reflecting their linear spatial structure that places some water pixels within close proximity to coastal industrial threats while others remain remote. Cropland exhibited lower within-class variability ($CV = 7.6\%$), consistent with its predominantly agricultural matrix context where threat exposure is more spatially homogeneous (Fig. 2).

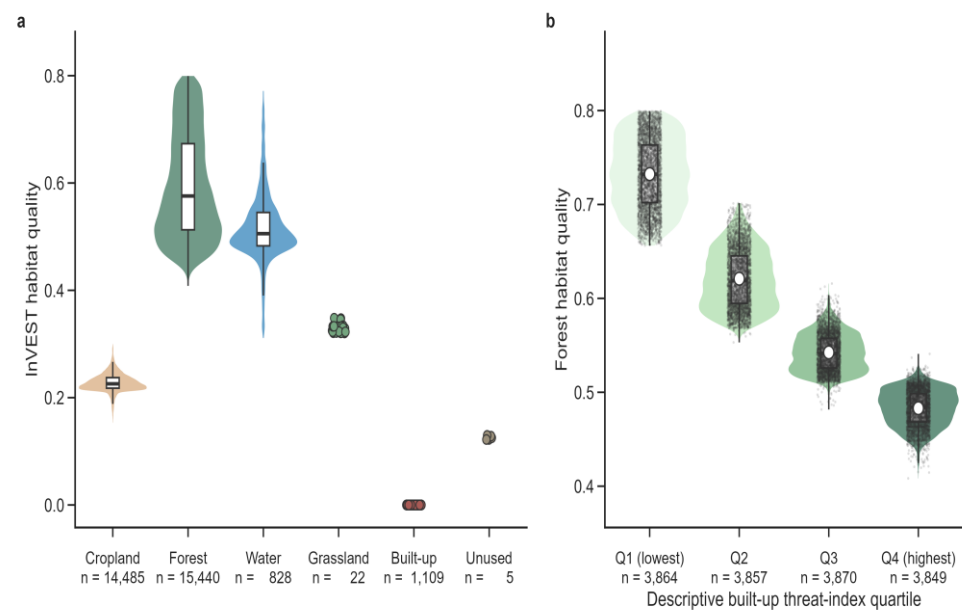


Fig. 2. Per-class distribution of InVEST habitat quality (HQ, 2020, 1-km grid) and descriptive forest stratification. (a) Raincloud/violin + raw-point distributions of HQ for six land-cover classes ($n = 31,889$; forest $n = 15,440$, cropland $n = 14,485$, water $n = 828$, built-up $n = 1,109$, grassland $n = 22$, unused $n = 5$). (b) Forest HQ by descriptive quartile of built-up threat-index exposure (identical $H_j = 0.80$; $n = 15,440$; Q1 $n = 3,864$, mean = 0.732; Q2 $n = 3,857$, mean = 0.621; Q3 $n = 3,870$, mean = 0.542; Q4 $n = 3,849$, mean = 0.483). Quartile comparisons are descriptive associations of spatial configuration, not fitted causal effects.

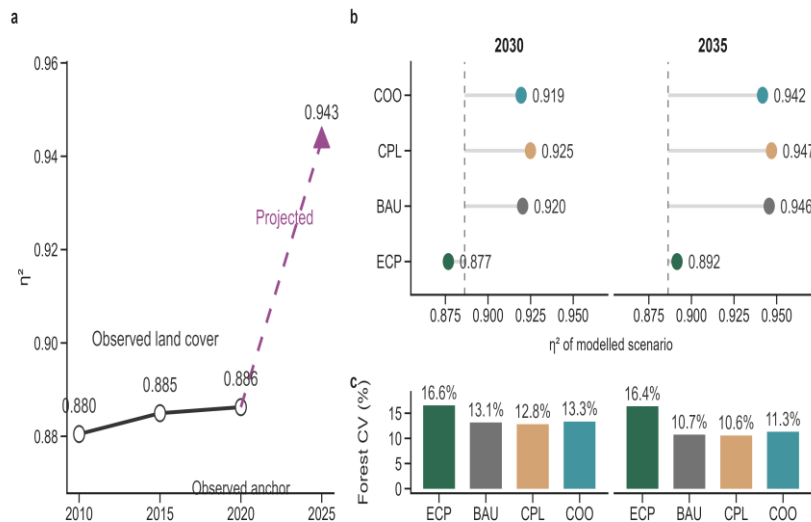


Fig. 3. Multi-temporal variance decomposition (η^2) of InVEST HQ. (a) Observed land cover for 2010, 2015, 2020 ($\eta^2 = 0.880, 0.885, 0.886$; $n = 31,889$ per year) and projected 2025 ($\eta^2 = 0.943$). No confidence bands are shown because each year is a single model output. (b, c) Four modelled scenarios (ECP, BAU, CPL, COO) at 2030 and 2035: scenario η^2 (b) and forest within-class CV (c). Baseline 2020 observed $\eta^2 = 0.886$. Scenarios are policy-sensitivity explorations, not winner identification.

4.2 Per-Class Within-Class Heterogeneity

To visualize the asymmetry between variance components across land cover types, we examined the distribution of HQ values within each class (Fig. 2). The box plots reveal marked differences in both central tendency and dispersion among the six land cover types, reflecting the combined effect of the habitat-suitability scores and the spatial heterogeneity of threat exposure.

The magnitude of within-class variation differed substantially across land cover types (Table 6), reflecting the interaction between each type's habitat suitability and the spatial heterogeneity of threat exposure (Fig. 2).

Forest exhibited the highest within-class CV (16.4%), indicating that nominally identical forest pixels experience substantially different habitat quality depending on spatial context. Forest pixels in the lowest built-up threat quartile had a mean HQ of 0.732 ($n = 3,864$), while those in the highest quartile had a mean HQ of 0.483 ($n = 3,849$), a difference of 0.249, representing 41.9% of the forest class mean. This

confirms that even within a single land cover class, neighborhood threat exposure creates ecologically meaningful HQ variation. (Fig. 2b).

Table 5. Multi-temporal η^2 comparison (2010-2025). $\Delta\eta^2 = 0.006$ over 2010-2020.

Year	n	Mean HQ	η^2	Between (%)	Within (%)	Forest CV (%)
2010	31,889	0.4142	0.881	88.1%	11.9%	16.7%
2015	31,889	0.4087	0.885	88.5%	11.5%	16.5%
2020	31,889	0.4052	0.886	88.6%	11.4%	16.4%
2025	31,889	0.4328	0.943	94.3%	5.7%	11.4%

These quartile-based comparisons are complemented by the full per-class statistical summary (Table 6), which reports the sample size, mean, standard deviation, coefficient of variation, and contribution to the total within-class sum of squares for each land cover type.

Table 6. Per-class within-class HQ statistics (2020). Forest and cropland together account for 95.3% of all within-class variance. Built-up land (HS = 0) produces identically zero HQ regardless of spatial context. Classes with extremely small sample sizes (Unused, n = 5; Grassland, n = 22) were retained for structural completeness but contribute negligibly (<0.1%) to the global within-class variance decomposition.

LUCC Class	Hj	n	Mean HQ	SD	CV (%)	% of SSW
Forest	0.8	15,440	0.595	0.0974	16.4%	94.8%
Cropland	0.3	14,485	0.228	0.0173	7.6%	2.8%
Water	0.8	828	0.517	0.0665	12.9%	2.4%
Grassland	0.35	22	0.330	0.0084	2.5%	<0.1%
Unused	0.2	5	0.125	0.0032	2.6%	<0.1%
Built-up	0.0	1,109	0.000	—	—	0.0%

Variable	η^2	Between (%)	Within (%)	Forest CV (%)
HQ (Habitat Quality)	0.886	88.6%	11.4%	16.4%
DEM (Elevation)	0.211	21.0%	79.0%	108.1%
PRE (Precipitation, WY proxy)	0.012	1.2%	98.8%	16.2%

The decomposition was temporally stable. η^2 remained at 0.881-0.886 across 2010-2020 ($\Delta\eta^2 = 0.006$) and rose to 0.943 by 2025 as projected land-cover change sharpened

between-class contrasts, while the forest within-class CV held at 16.4-16.7% over 2010-2020 (Table 5; Fig. 3a).

4.3 Cross-Variable Comparison: η^2 as a Universal Diagnostic

To verify that the high η^2 of HQ is a property of the InVEST parameterization rather than a landscape property, we applied the same variance decomposition framework to two additional variables: DEM (elevation) and mean annual precipitation (PRE). The results (Table 7) were clear: PRE showed an η^2 of only 0.012 (1.2% between-class), demonstrating that when a variable is truly independent of land cover type, variance decomposition produces a near-zero η^2 . DEM showed an intermediate η^2 of 0.211. The HQ η^2 (0.886) was 74 times higher than the PRE η^2 , confirming that the dominant between-class structure of HQ is an artefact of InVEST's HS lookup-table parameterization. Within-class residual analysis further supported this: HQ residuals correlated strongly with DEM residuals ($r = 0.65$), consistent with the topography-to-threat-configuration mediated pathway, but only weakly with PRE residuals ($r = 0.10$).

Table 7. Cross-variable η^2 comparison. HQ η^2 is $74 \times$ PRE η^2 .

4.4 Scenario-Dependence of the Variance Decomposition

The multi-scenario analysis (Table 8; Fig. 3b,c) demonstrated that η^2 is sensitive to policy-driven land use pathways. Under the Ecological Protection (ECP) scenario, η^2 remained at 0.877-0.892 (2030-2035), with forest CV preserved at 16.4-16.6%. Under the Cropland Protection (CPL) scenario, η^2 rose to 0.925-0.947, and forest CV collapsed to 10.6-12.8%. The η^2 gap between ECP and CPL widened from 0.048 (2030) to 0.056 (2035), indicating that policy divergence amplifies the erosion of spatial configuration diversity over time.

This finding demonstrates that η^2 is not merely a static diagnostic but a policy-sensitive indicator.

Table 8. Multi-scenario η^2 comparison. $\Delta\eta^2(\text{ECP-CPL}) = 0.048$ to 0.056 .

Scenario	Year	η^2	Within (%)	Forest CV (%)	Built-up (n)
BASELINE	2020	0.8863	11.4%	16.4%	1,109
ECP	2030	0.8767	12.3%	16.6%	1,266

BAU	2030	0.9203	8.0%	13.2%	1,370
CPL	2030	0.9249	7.5%	12.8%	1,283
COO	2030	0.9194	8.1%	13.3%	1,440
ECP	2035	0.8915	10.8%	16.4%	1,402
BAU	2035	0.9456	5.4%	10.7%	1,444
CPL	2035	0.9470	5.3%	10.6%	1,400
COO	2035	0.9417	5.8%	11.3%	1,505

4.5 Two-Stage Geodetector: Disentangling Exogenous Drivers from Land-Cover

Mediation

After LUCC detrending, slope collapsed to $q_{\text{resid}} = 0.147$, retaining only 28.4% of its apparent power, whereas elevation ($q_{\text{resid}} = 0.309$, 64.8% retention) and temperature ($q_{\text{resid}} = 0.292$, 62.9%) retained the highest residual explanatory power. This contrast, high raw q but low retention for slope versus high retention for elevation and temperature, is the empirical signature of the land-cover confounding that the framework quantifies: slope influences HQ predominantly by constraining where land cover and its associated threat sources can occur, not through a direct topographic effect, while elevation and temperature retain explanatory power beyond land-cover mediation.

4.6 Robustness Verification

The ranking was robust to discretization: under equal-interval strata, slope retention stayed near one-third and elevation and temperature retention exceeded 60% (Table 9). The interaction detector showed that all leading factor pairs exhibited bifactor enhancement. The strongest interactions were slope $\times$ DEM ($q = 0.580$) and slope $\times$ NDVI ($q = 0.574$), indicating that topographic gradients amplify other factors' explanatory power without antagonistic interaction.

Table 9. Two-stage Geodetector q -statistics for InVEST HQ driver attribution (complete-case $N = 30,711$; 5-quantile (quintile) strata, equal-interval in parentheses as a robustness check; Yuexi 2020). Retention = $q_{\text{resid}} / q_{\text{raw}}$ under 5-quantile (quintile) strata.

Factor	q_{raw}	q_{resid}	Retention %	q_{raw} (EI)	q_{resid} (EI)
Slope	0.517	0.147	28.4	0.464	0.153

Elevation (DEM)	0.476	0.309	64.8	0.258	0.299
Temperature	0.464	0.292	62.9	0.378	0.321
Soil type	0.382	0.259	67.9	0.382	0.259
NDVI	0.317	0.150	47.3	0.227	0.058
Nighttime light	0.235	0.082	34.9	0.160	0.034
Population	0.133	0.082	61.8	0.014	0.000
Dist. to roads	0.111	0.076	68.6	0.108	0.115
GDP	0.086	0.047	55.0	0.013	0.000
Dist. to railways	0.045	0.031	69.1	0.032	0.027
Dist. to water	0.040	0.026	64.4	0.036	0.024
Precipitation	0.022	0.014	64.7	0.036	0.015
Aspect	0.008	0.001	9.6	0.008	0.001

4.7 Spatial Clustering of Forest Within-Class Residuals

Local Moran's I applied to the forest within-class HQ residuals yielded a global Moran's I of 0.752 ($p < 0.001$; $n = 15,440$), confirming strong spatial autocorrelation in the within-class component. Significant high-high (HH) clusters accounted for 24.0% of forest points and concentrated in the northern Tianlu-Yunwu forest core, whereas low-low (LL) clusters accounted for 28.9% and aligned with the coastal urban corridor; high-low and low-high spatial outliers were rare (1.0% and 0.2%, respectively), and 45.8% of points were not statistically significant (Fig. 4).

4.8 Policy Implications

The variance decomposition has direct policy relevance. Conservation planning traditionally focuses on “how much” of each land cover type to protect. Within the InVEST framework, 88.6% of HQ outcomes are already determined by aggregate land cover proportions. The remaining 11.4%, the neighborhood configuration component, is where spatial planning can meaningfully improve outcomes for a fixed land cover composition (Lü et al., 2012).

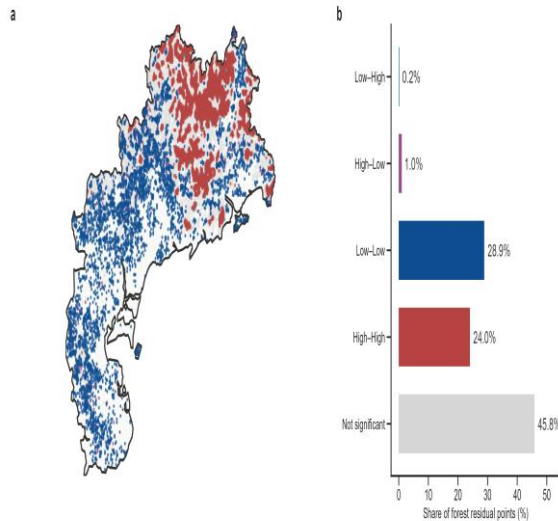


Fig. 4. Local Moran's I (LISA) clusters of forest within-class HQ residuals (2020; $n = 15,440$ forest sampling points; KNN $k = 8$; 999 permutations; $p < 0.05$). Global Moran's $I = 0.752$. (a) LISA cluster map. (b) Cluster shares: high-high 24.0%, low-low 28.9%, high-low 1.0%, low-high 0.2%, non-significant 45.8%.

Three independent checks confirmed the stability of the decomposition (Tables 10, 11; Fig. 5). The Monte Carlo perturbation of HS ($n = 200$, $\pm 20\%$) produced η^2 values that never fell below 0.826 (mean = 0.883), so the 88.6% structural dominance is not an artifact of the specific sensitivity-table scores (Fig. 5c). Spatial thinning from 1 km to 10 km minimum inter-point spacing left η^2 essentially unchanged (0.88-0.90) while the global Moran's I of HQ declined from the full-set value 0.615 to 0.406 at 10 km (1 km thinned value = 0.568), indicating that η^2 is not inflated by spatial autocorrelation (Table 10; Fig. 5a,b). The k -sensitivity analysis showed that η^2 rises monotonically with the half-saturation constant, from 0.144 at $k = 0.1$ to 1.000 at $k \geq 2.0$ (Table 11; Fig. 5d); changing k from 0.5 to the literature-reported value of 2.8 (Wu et al., 2021; Pan et al., 2024) shifts η^2 by 0.114, approximately 20 times the change observed over a decade of landscape evolution ($\Delta\eta^2 = 0.006$ between 2010 and 2020; Table 5; Fig. 3a). We further note that Two-stage Geodetector significance tests are subject to spatial autocorrelation (residual Moran's $I = 0.752$). Recomputing the full decomposition and two-stage Geodetector at the half-maximum calibration guideline ($k = 0.30$) confirmed that the driver hierarchy is robust to this choice: elevation and temperature retained the top residual ranks ($q = 0.309$ and 0.292 , complete-case $n = 30,711$), whereas the between-class share of variance declined from 0.886 to 0.586 (Table 11); the driver hierarchy is

robust to k while the absolute effect-size magnitudes are not. The spatial thinning experiment (Section 4.7) confirms η^2 stability, but future applications should consider spatially constrained permutation tests. The ranking of residual driver importance was also robust to the discretization of continuous predictors: across four discretization methods (quantile, equal-interval, Jenks, and standard-deviation) and three bin counts (5, 8, and 10), elevation and temperature ranked first or second in all 12 parameter combinations, whereas no other factor entered the top two. As a further cross-check, the driver hierarchy was independently examined with machine-learning attribution: an XGBoost model of HQ on the thirteen exogenous factors (five-fold CV $R^2 = 0.683$, SD 0.004 across folds) yielded TreeSHAP importances in which slope and DEM ranked first and second (Fig. 6c), consistent with the two-stage Geodetector result that these factors retain the strongest residual explanatory power.

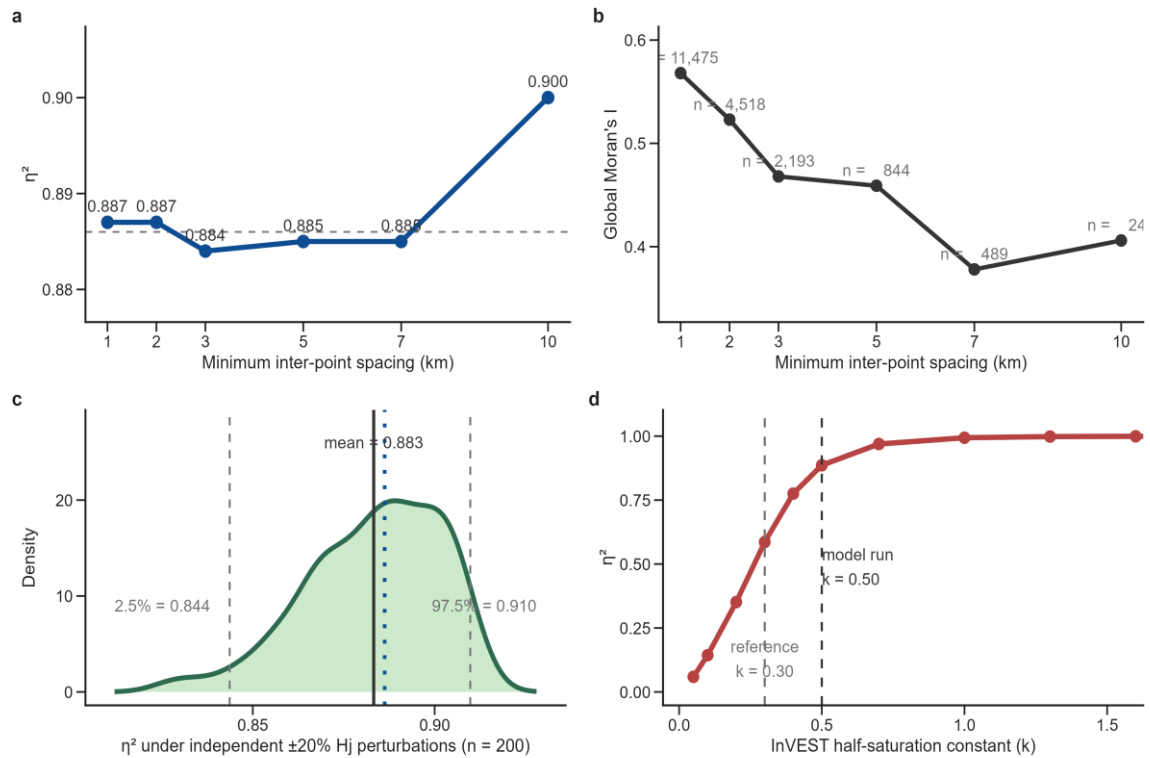


Fig. 5. Sensitivity diagnostics of the variance decomposition. (a) η^2 across greedy spatial thinning (1–10 km minimum inter-point spacing); η^2 remained within 0.884–0.900. (b) Global Moran's I of HQ declined from the full-set value 0.615 to 0.406 at 10 km (first thinned spacing 1 km = 0.568); thinning values originate from the manuscript's Table 10 (n = 241–11,475 retained), whose historical producer has not been independently reproduced. (c) η^2 under independent $\pm 20\%$ H_j perturbations (n = 200

Monte Carlo; mean = 0.883, range [0.826, 0.914]); the dotted vertical line marks the observed 2020 $\eta^2 = 0.886$. (d) η^2 vs half-saturation constant k ($z = 2.5$ fixed); x-axis is tightened to the sensitive range [0.02, 1.55] — η^2 exceeds 0.998 for $k \geq 1.3$; $k = 0.30$ is a reference calibration, $k = 0.50$ is the model run; no optimal- k claim is intended.

5. Discussion

The η^2 of 0.886 has direct implications for the interpretation of Geodetector results, as it confirms that the majority of HQ variance is structural rather than stochastic. The collapse of the slope driver from $q = 0.517$ to $q = 0.147$ illustrates how much apparent driver power in conventional analyses is attributable to land-cover confounding. the persistence of elevation and temperature as the factors retaining the strongest residual explanatory power suggests that these topographic and climatic factors provide the fundamental “scaffolding” for habitat quality, even when land cover types vary.

Our analysis reveals that while land cover provides the primary identity of HQ, the within-class configuration (e.g., forest's CV of 16.4%) determines the local nuances of habitat suitability. The temporal stability of η^2 (0.881–0.943) across 2010–2025 highlights the robustness of our variance decomposition. The k -sensitivity analysis highlights the importance of parameter selection, showing that the variance structure is highly sensitive to the k value, which can be tuned to balance local and global variance components.

In practical terms, the audit-and-sensitivity workflow allows researchers to move beyond simple “dominant factor” claims and instead quantify the association retained by exogenous variables after land-cover adjustment. By accounting for land-cover-aligned structure, we can quantify the residual association of topographic and climatic factors, which are often masked by the HS lookup table. This study provides a sensitivity workflow for assessing habitat quality and its drivers in both natural and anthropogenically modified landscapes.

5.1 Implications for InVEST-Based Driver Attribution Studies

The η^2 of 0.886 has direct implications for the interpretation of Geodetector analyses applied to InVEST HQ outputs. When a Geodetector model identifies LUCC

as the “dominant driver” of HQ ($q \sim 0.85\text{--}0.90$ in our baseline runs and widely reported in the literature), it is simply recovering the parameterization that was programmed into InVEST: $H_{\text{cropland}} \neq H_{\text{forest}} \neq H_{\text{water}}$ (Chen et al., 2023; He et al., 2019; Liu et al., 2020; Liu et al., 2024; Yesuph and Dagnew, 2019).

The apparent “explanatory power” of topographic variables (DEM, slope) in raw Geodetector analyses reflects a mediated pathway: topography constrains where anthropogenic threat sources (especially built-up land) are located, which in turn determines neighborhood threat exposure for nearby pixels of the same land cover type. Our two-stage Geodetector makes this mediation quantitative: once HQ is detrended for land-cover class means, the apparent dominance of slope collapses from $q = 0.517$ to $q = 0.147$, retaining only 28% of its raw explanatory power, whereas elevation ($0.476 \rightarrow 0.309$) and temperature ($0.464 \rightarrow 0.292$) retain 64.8% and 62.9%, respectively and emerge as the factors with the highest residual retention.

The within-class HQ residual correlation with DEM ($r = 0.65$) versus the near-zero correlation with precipitation ($r = 0.10$) independently corroborates this ranking. Slope's apparent influence is thus largely a spatial-configuration artefact operating within the 11.4% within-class component, not a direct topographic control on habitat quality.

We therefore recommend that future InVEST-Geodetector studies: (1) report η^2 as a diagnostic to quantify the dominance of the structural effect; (2) disaggregate LUCC from the exogenous variable set rather than reporting it as a “driver”; and (3) apply a two-stage Geodetector that recomputes q -statistics on within-class (LUCC-detrended) HQ residuals, so that drivers retaining explanatory power after adjustment are separated from those whose apparent influence is largely aligned with land-cover distribution. This protocol is conceptually related to variation partitioning in ecology (Borcard et al., 1992; Peres-Neto et al., 2006). Because attribution applied to raw InVEST outputs can inherit the land-cover confounding documented here, the diagnostic-and-adjustment sequence can serve as a preprocessing step before subsequent attribution analyses.

Concretely, the 0.249 HQ difference between lowest and highest threat-exposure quartiles represents a theoretical upper bound for spatial configuration effects within the

InVEST model framework, rather than a guaranteed ecological gain. This value quantifies the maximum within-class variance attributable to threat-source proximity under the Euclidean distance assumption. Policy interventions targeting spatial configuration should be evaluated by their ability to mitigate the model-identified configuration risk rather than by absolute InVEST HQ gains. Under the ECP scenario, where spatial configuration is partially preserved, this risk is maintained rather than amplified, supporting the value of spatially-aware land-use planning.

Table 10. Spatial thinning verification of η^2 .

Spacing (m)	N retained	Moran's I	η^2	$\Delta\eta^2$
1,000	11,475	0.568	0.887	+0.001
2,000	4,518	0.523	0.887	+0.001
3,000	2,193	0.468	0.884	-0.002
5,000	844	0.459	0.885	-0.001
7,000	489	0.378	0.885	-0.001
10,000	241	0.406	0.900	+0.014
Full set	31,219	0.615	0.886	—

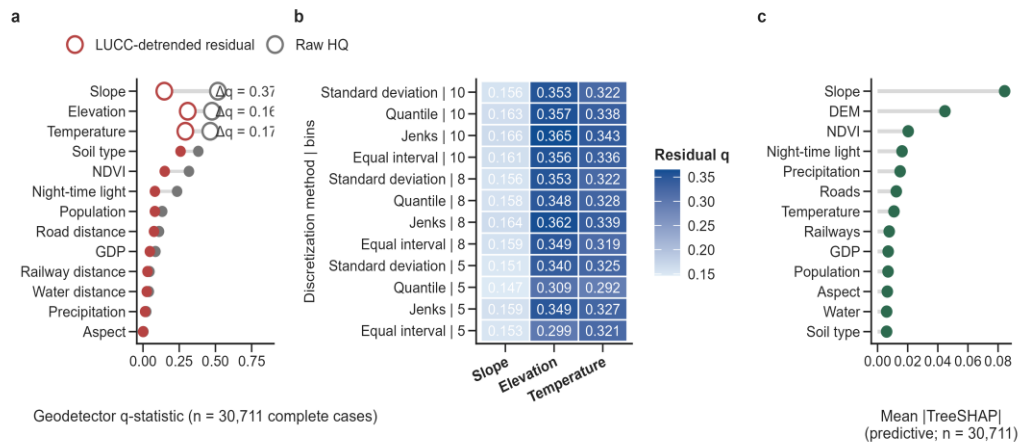


Fig. 6. Driver attribution of HQ variance and sensitivity to land-cover detrending. (a) Paired raw-to-residual Geodetector q-statistics for 13 factors ($n = 30,711$ complete cases) before and after LUCC detrending; top-3 attenuation Δq : Slope 0.370, Elevation 0.167, Temperature 0.172. (b) Residual-q sensitivity across four discretization methods (quantile, equal-interval, Jenks, standard-deviation) $\times$ three bin counts (5, 8, 10) for the three leading factors. (c) Mean |TreeSHAP| from the XGBoost model of HQ on thirteen exogenous factors (five-fold CV $R^2 = 0.683$); a predictive-attribution artifact, not a causal statement. The spatial-share attribution output is excluded because its denominator could not be reconciled with the stored artifact.

513 Table 11. k-value sensitivity. k effect on the decomposition ($\Delta\eta^2$).

k	η^2	Between-class %	Within-class %
0.1	0.144	14.4	85.6
0.30	0.586	58.6	41.4
0.5	0.886	88.6	11.4
1.0	0.994	99.4	0.6
1.5	0.999	99.9	0.1
2.0	1.000	~100	<0.1
2.5	1.000	~100	<0.1
2.8	1.000	~100	<0.1
5.0	1.000	~100	<0.1

514 5.2 Interpreting the Within-Class Component

515 The decomposition proved robust to all three independent checks, HS perturbation,
516 spatial thinning, and k-variation (Section 4.7). The remaining within-class component,
517 however, must be interpreted with care.

518 Euclidean Distance Reductionism. We caution that the 11.4% within-class variance
519 component should not be naively equated with “ecological neighborhood effects.” The
520 InVEST model computes threat degradation using isotropic Euclidean distance
521 transforms that ignore terrain barriers, prevailing wind direction, and hydrological
522 connectivity. In mountainous terrain, two forest pixels at identical Euclidean distances
523 from a road may experience fundamentally different ecological stress. The “spatial
524 configuration effect” quantified here is a Model Configuration Effect, an upper bound
525 on potential improvement achievable through spatial planning within the InVEST
526 framework. We recommend that future studies replace Euclidean distance decay with
527 least-cost path distance based on MCR (Minimum Cumulative Resistance) models that
528 incorporate topographic impedance (Rao et al., 2019; Peng et al., 2019).

529 5.3 Limitations

530 Our workflow is specific to the InVEST HQ formulation and the Yuexi
531 parameterization. Different threat-source configurations (e.g., adding roads as threat

sources), different half-saturation constants, or landscapes with more heterogeneous threat distributions would alter the between/within split. The two-stage adjustment also assumes that land-cover class means capture the dominant structural signal and that the residual q-statistics are interpreted as net-exogenous associations, rather than causal effects. Nonetheless, the methodological principle, diagnosing structural dominance by variance decomposition and then recomputing driver attribution on detrended residuals, is broadly applicable; our workflow should be understood as a sensitivity template, not a fixed partitioning.

6. Conclusions

This study presents an audit-and-sensitivity workflow for InVEST Habitat Quality (HQ) analysis, providing a method to separate land-cover-aligned structure from exogenous variables. By decomposing the variance structure, we demonstrated that 88.6% of HQ variance is between-class, while the remaining 11.4% captures the nuances of within-class spatial configuration. Our two-stage Geodetector adjustment quantified how much of each factor's apparent explanatory power survives land-cover detrending, revealing that elevation and temperature retain the strongest residual associations with HQ heterogeneity.

The findings offer three recommendations: (1) report η^2 to quantify the structural dominance of land cover; (2) use land-cover-detrended residuals to quantify the residual explanatory power of topographic variables; and (3) identify configuration-sensitive classes (e.g., forest) to prioritize local habitat conservation. The workflow provides a sensitivity protocol for assessing habitat quality in LULC-parameterized InVEST outputs.

Data Availability

All spatial point datasets, regression models, R/Python analytical scripts, and InVEST model parameter files supporting this study are available in the open repository at Zenodo (DOI: 10.5281/zenodo.XXXXXXX) and GitHub (https://github.com/shi dikuangsan31/yuexi_hq_audit). Full computational provenance for the analysis pipeline is archived with verified SHA-256 hashes of

input rasters, intermediate spatial point tables (n = 31,889 full domain; n = 30,711 complete cases with projected planar coordinates), official InVEST outputs, and run logs.

Conflict of Interest

The authors declare no conflict of interest.

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